Recommended Assessment

Infrared Distance

Exploring Infrared Distance Data

1. What do the raw voltage measurement from the Infrared Distance sensor tell you about how the sensor behaves?

2. What do the minimum and maximum values tell us about the range of the Infrared distance sensor?

3. How to the calibrated A and b values match the sampled data?

4. Using the calibrated A and b values, does the reported distance measurement match the calibration ruler? What other considerations might you need to take when using an infrared distance sensor?

Real Life Scenarios

5. You're design a series of mobile robots which use Infrared distance sensors to detect the presence of objects as obstacles. You notice there are random instances when the mobile robot enters an obstacle detection routine. This effect becomes worse when you go outdoors. What are some design considerations for the Infrared distance sensor which could be causing this issue?

6. You're using an Infrared distance sensor to count objects passing through a conveyor belt. You've noticed that metallic objects are sometimes counted incorrectly. What are some design changes you can make to increase the reliability of the object counter?

Infrared Distance Cheat Sheet

What is it?	An emitter/receiver pair used to send a pulse of infrared energy, bounce it from an object and measure the response.
What does it measure directly?	Distance between sensor and object within the sensor field of view.
What can you measure indirectly?	Assuming the sensor is stationary, you can measure the speed at which the object is approaching the sensor, but this will be a noisy value.
Where are they used?	These sensors are used in mobile robots for obstacle avoidance, warehousing for counting objects passing through the field of view, security for detecting objects passing through a specific area.
Pros	Low powered, measures distance independent of lighting conditions, works well indoors.
Cons	Sensitive to Infrared radiation, very small measurement range, requires constant calibration to account for environmental changes, can suffer from crosstalk between different Infrared distance sensors.
Notes/ Important things to remember	Surface finishes can affect distance measurement quality; non-linear calibration must be performed prior to using sensor. Objects outside of the measurement range can give erroneous measurements.